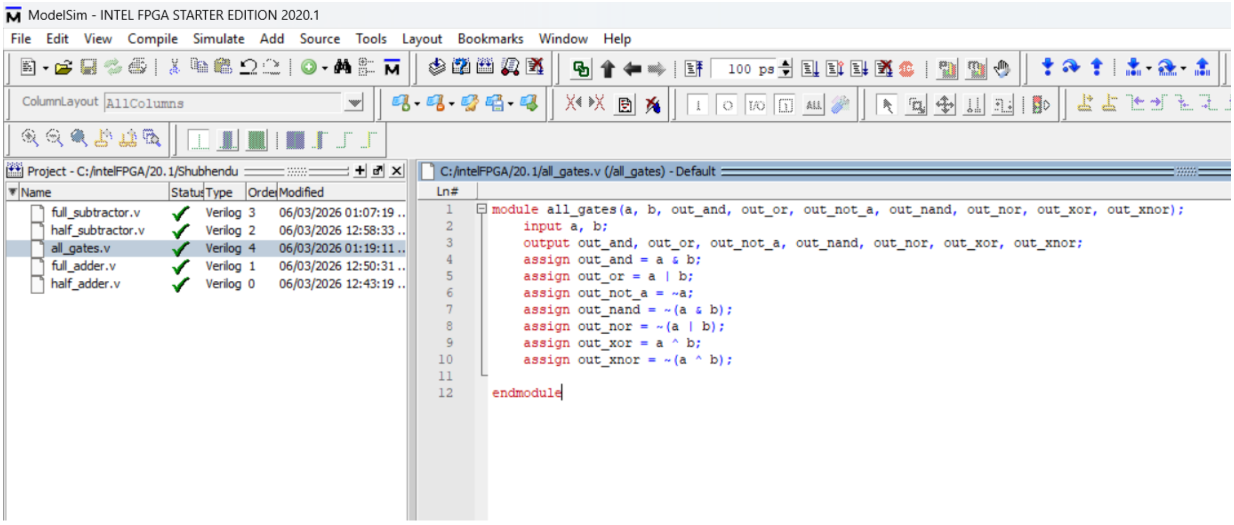
COMPUTER ARCHITECTURE LABORATORY MANUAL

DIGITAL LOGIC SIMULATION RECORDS

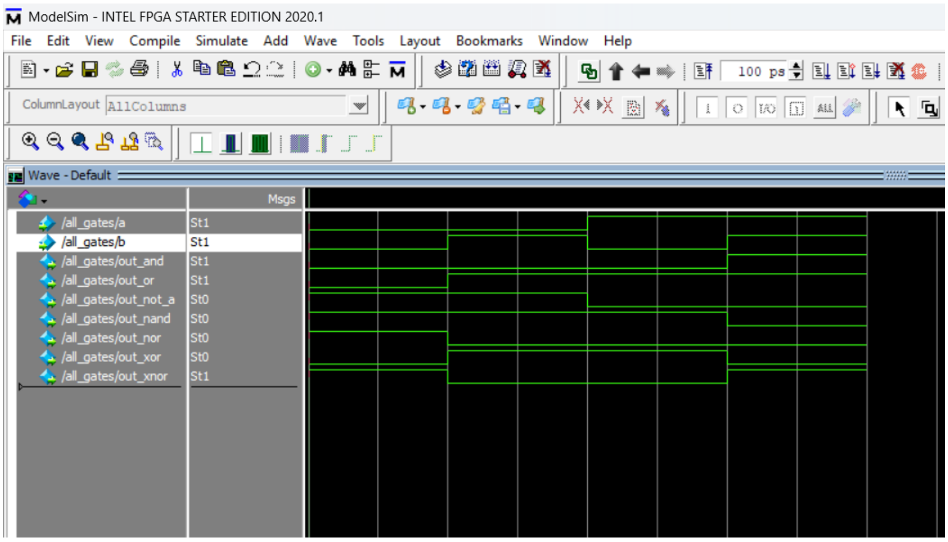
EXPERIMENT 1: Implementation of GATES

Objective: To design, simulate, and verify basic and universal logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) using Verilog HDL dataflow modeling.

--- Verilog Hardware Description Code ---



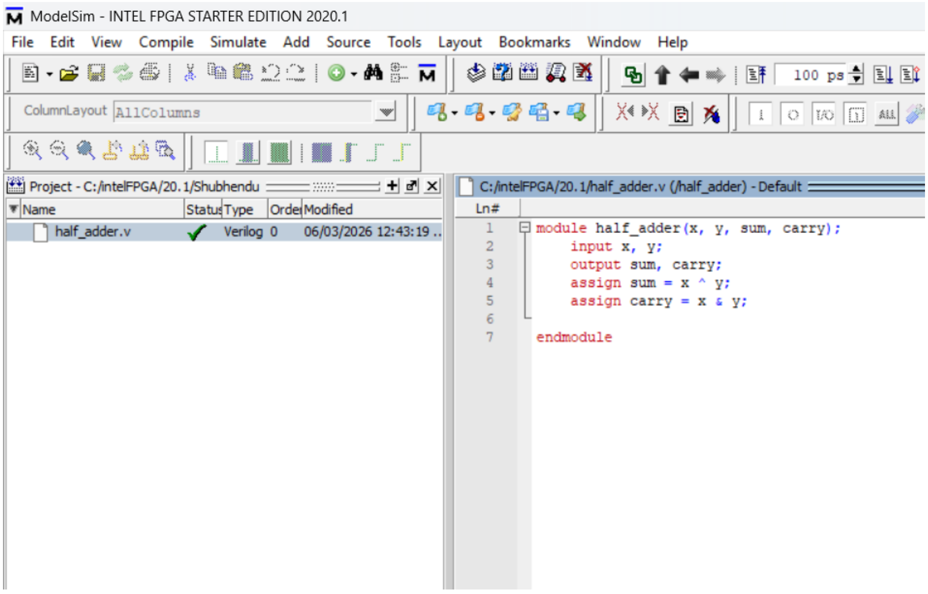
--- Simulation Waveform Output Trace ---



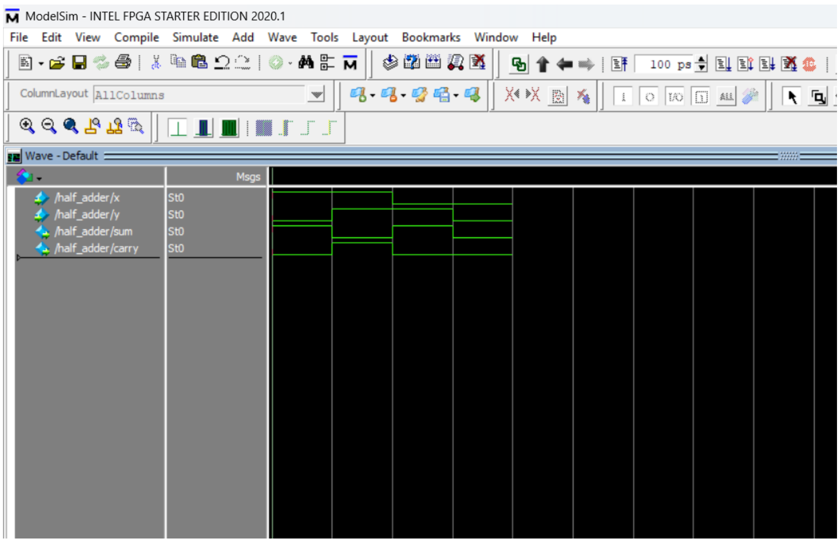
EXPERIMENT 2: Implementation of HALF ADDER

Objective: To design, simulate, and verify a 1-bit Half Adder using Verilog HDL dataflow modeling and analyze the Sum and Carry outputs.

--- Verilog Hardware Description Code ---



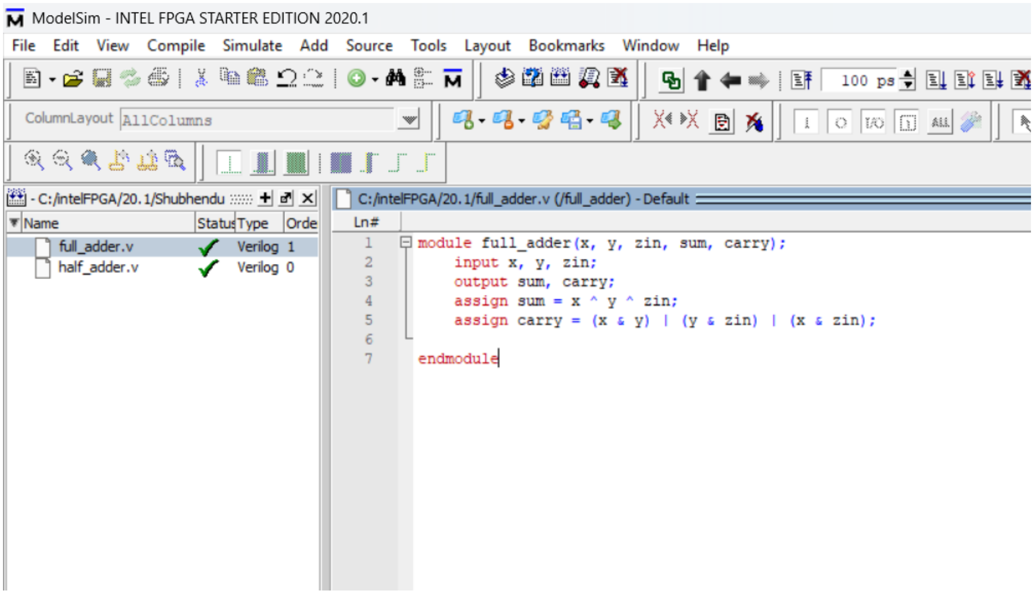
--- Simulation Waveform Output Trace ---



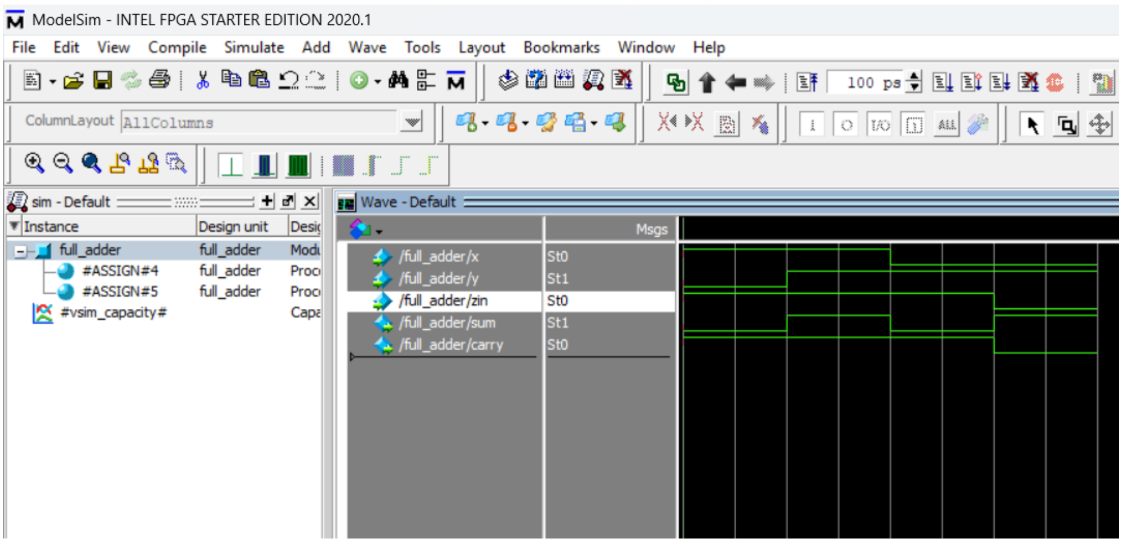
EXPERIMENT 3: Implementation of FULL ADDER

Objective: To design, simulate, and verify a 1-bit Full Adder using Verilog HDL dataflow modeling and analyze the Sum and Carry outputs with Carry-in input.

--- Verilog Hardware Description Code ---



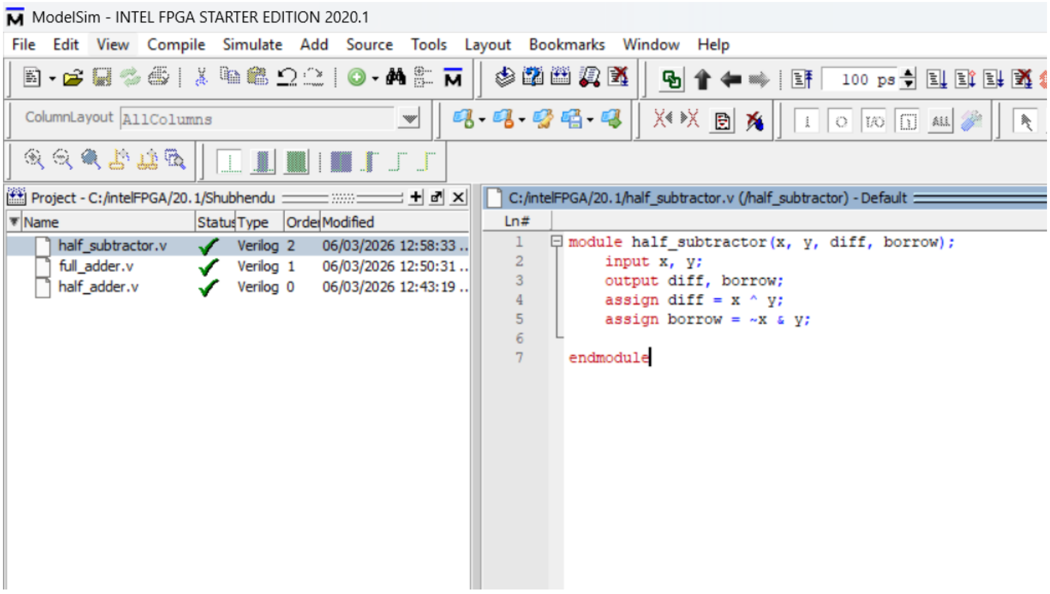
--- Simulation Waveform Output Trace ---



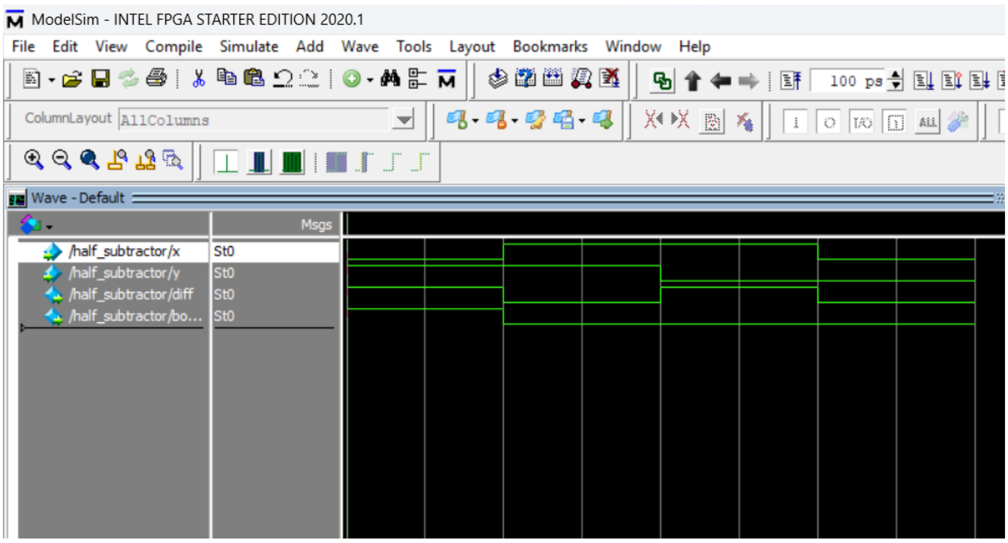
EXPERIMENT 4: Implementation of HALF SUBTRACTOR

Objective: To design, simulate, and verify a 1-bit Half Subtractor using Verilog HDL dataflow modeling and analyze the Difference and Borrow outputs.

`--- Verilog Hardware Description Code ---`



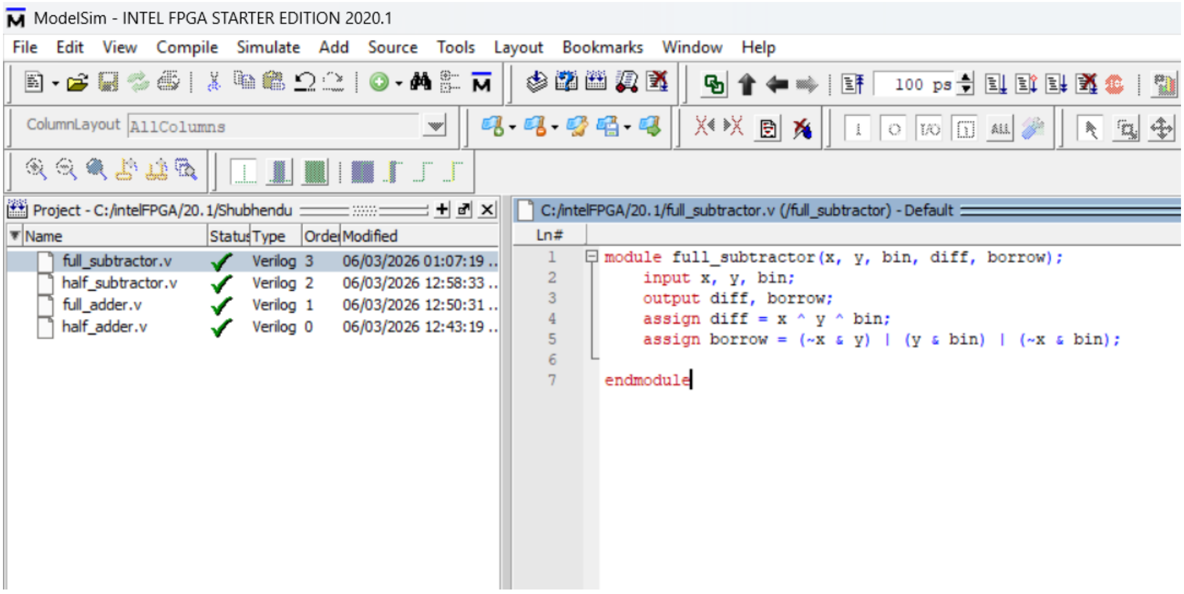
`--- Simulation Waveform Output Trace ---`



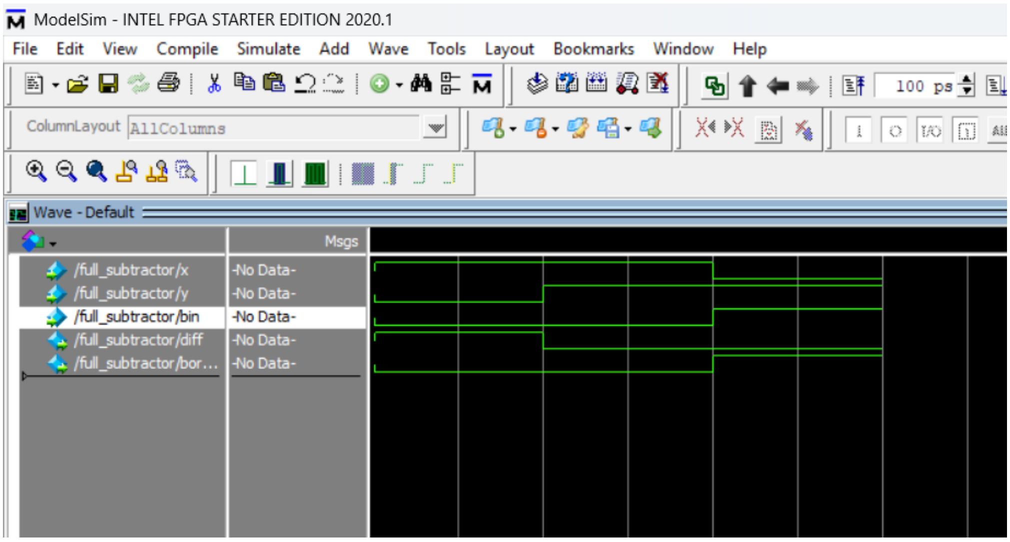
EXPERIMENT 5: Implementation of FULL SUBTRACTOR

Objective: To design, simulate, and verify a 1-bit Full Subtractor using Verilog HDL dataflow modeling and analyze the Difference and Borrow outputs with Borrow-in input.

`--- Verilog Hardware Description Code ---`



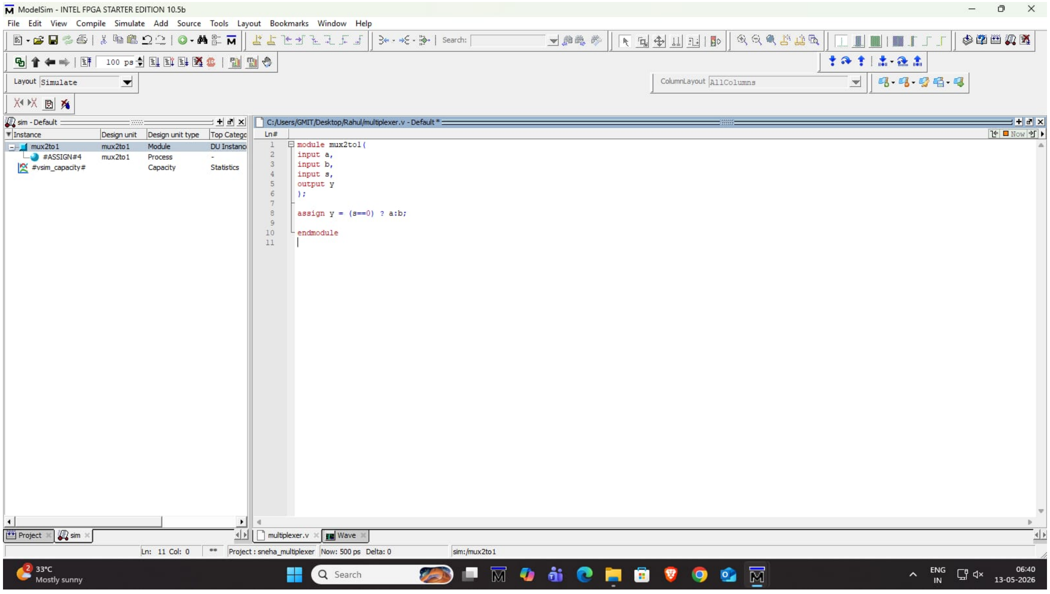
`--- Simulation Waveform Output Trace ---`



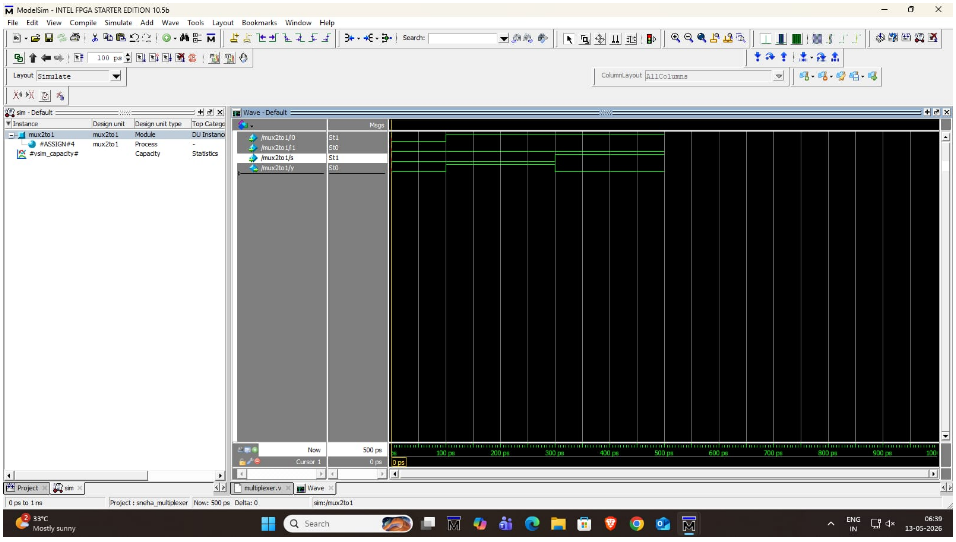
EXPERIMENT 6: Implementation of MUX 2X1

Objective: To design, simulate, and verify a 2x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through a single select line.

--- Verilog Hardware Description Code ---



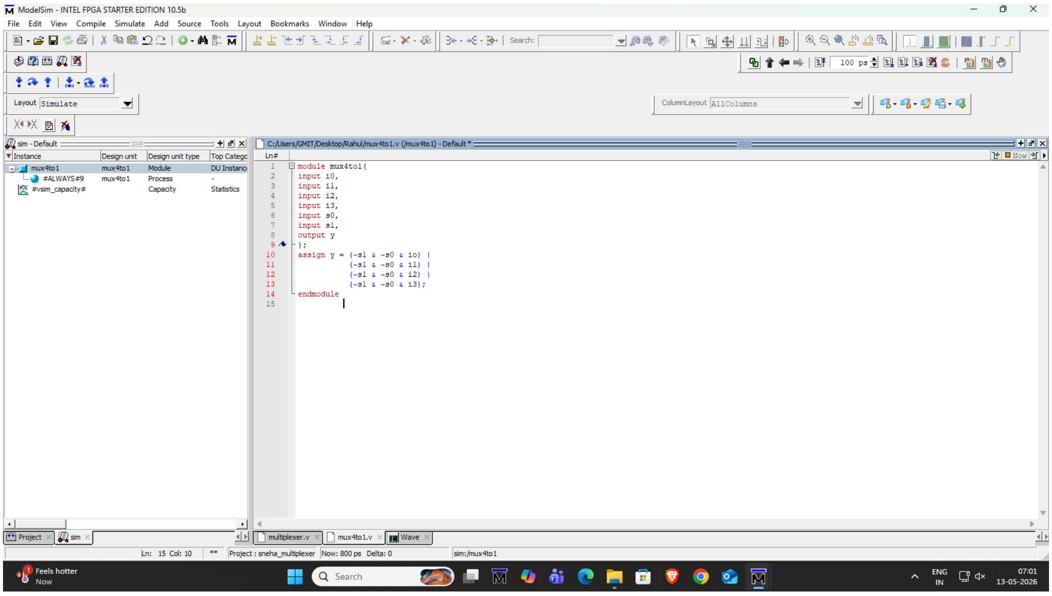
--- Simulation Waveform Output Trace ---



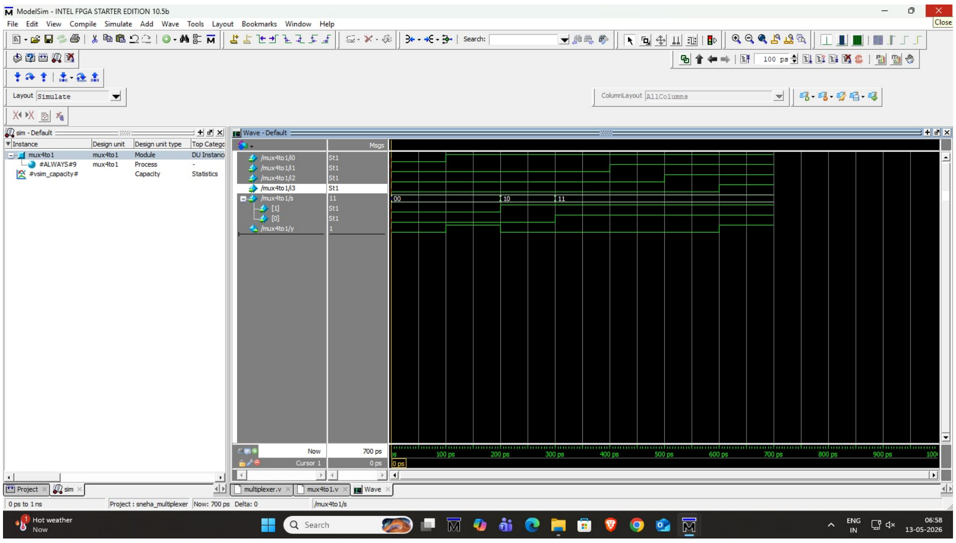
EXPERIMENT 7: Implementation of MUX 4X1

Objective: To design, simulate, and verify a 4x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through two select lines.

--- Verilog Hardware Description Code ---



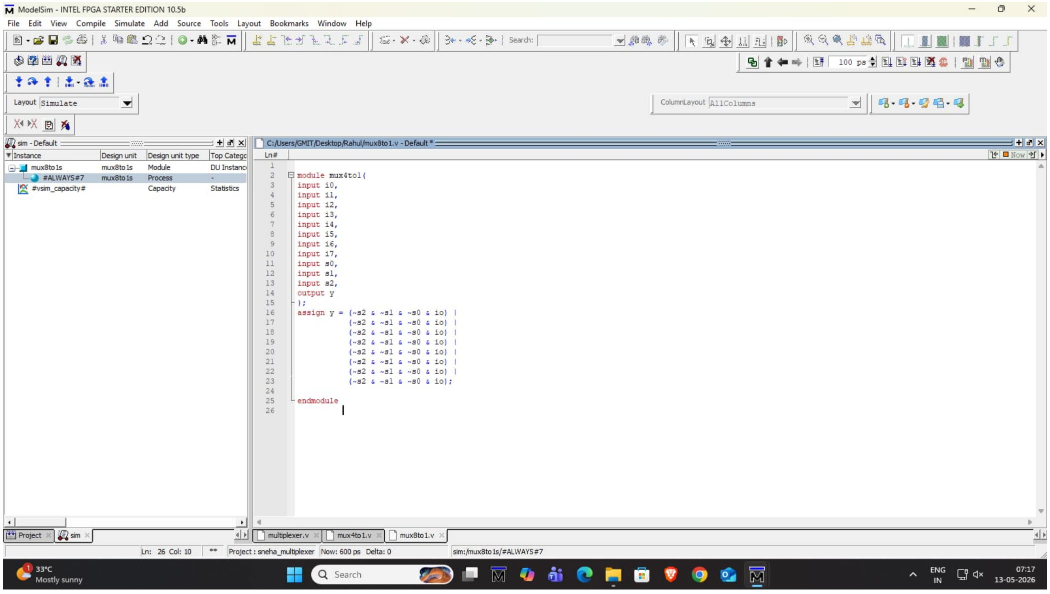
--- Simulation Waveform Output Trace ---



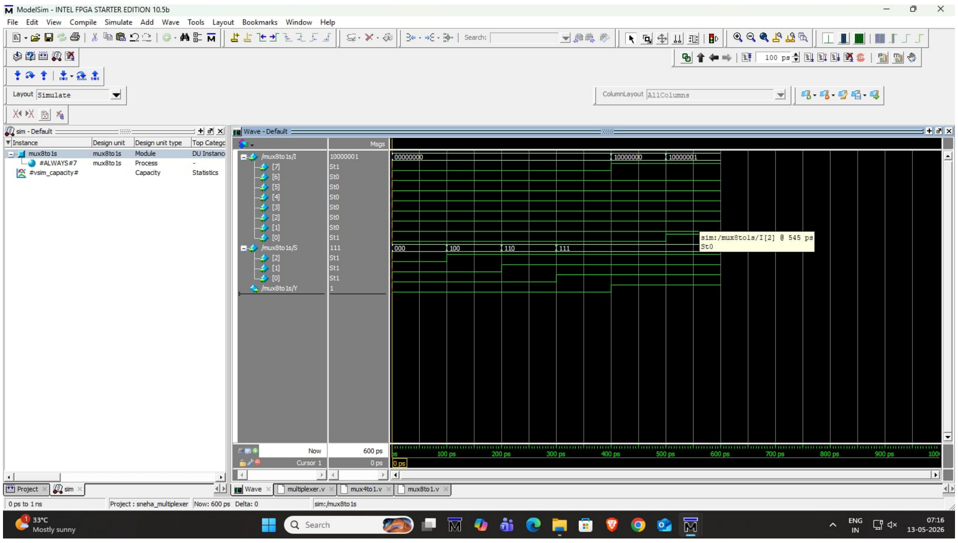
EXPERIMENT 8: Implementation of MUX 8X1

Objective: To design, simulate, and verify an 8x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through three select lines.

--- Verilog Hardware Description Code ---



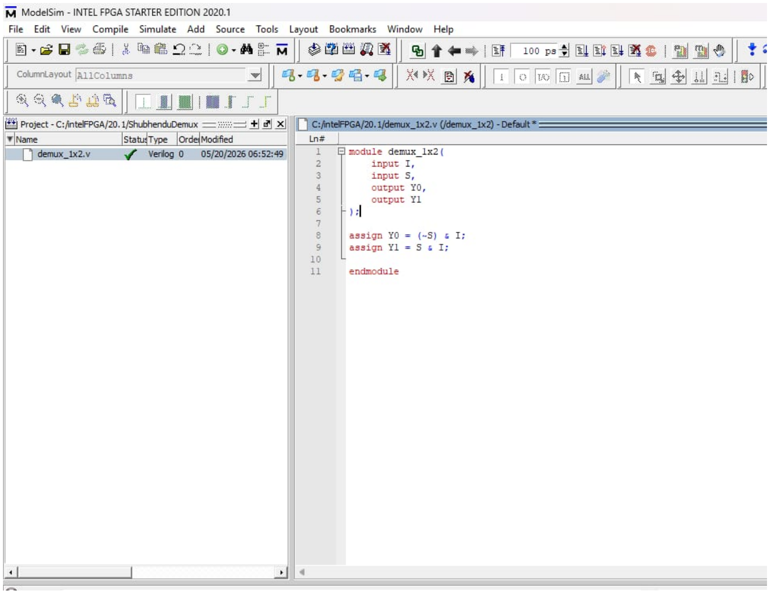
--- Simulation Waveform Output Trace ---



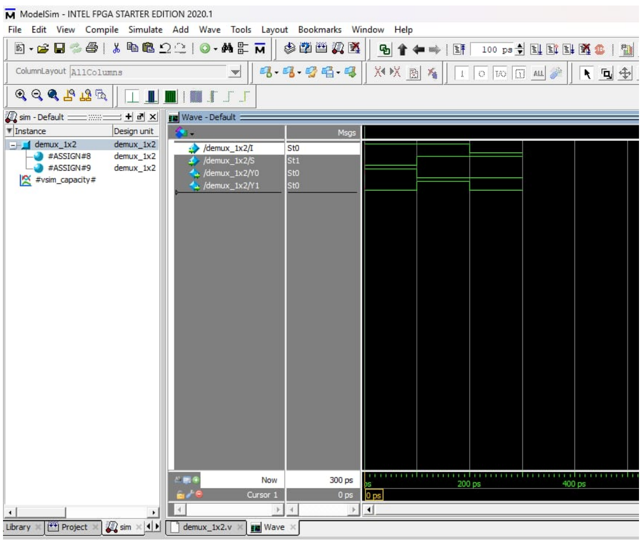
EXPERIMENT 9: Implementation of DEMUX 1X2

Objective: To design, simulate, and verify a 1x2 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of two outputs.

--- Verilog Hardware Description Code ---



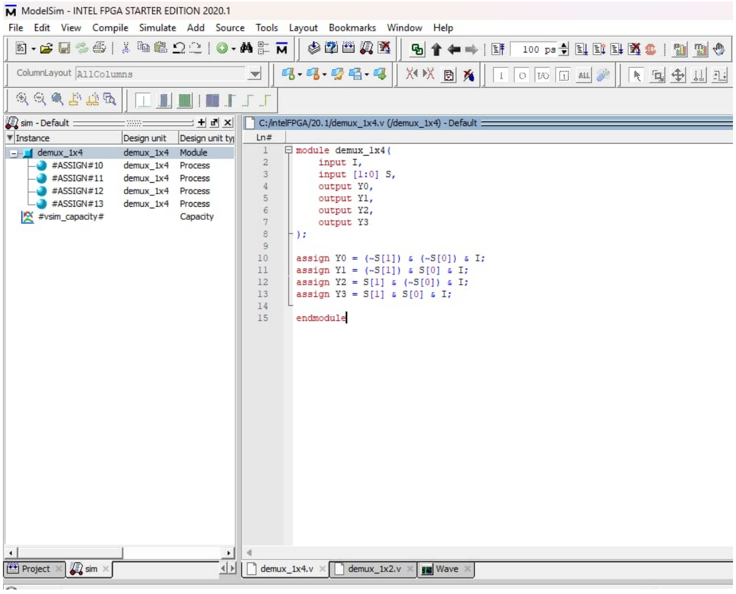
--- Simulation Waveform Output Trace ---



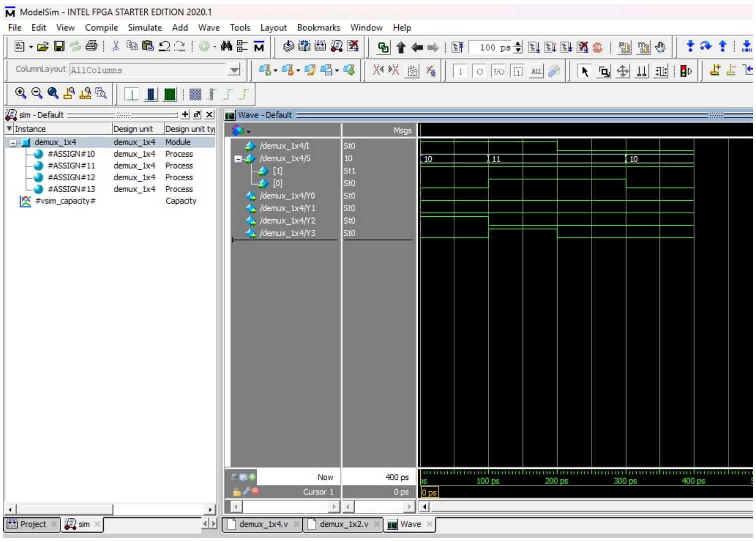
EXPERIMENT 10: Implementation of DEMUX 1X4

Objective: To design, simulate, and verify a 1x4 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of four outputs.

--- Verilog Hardware Description Code ---



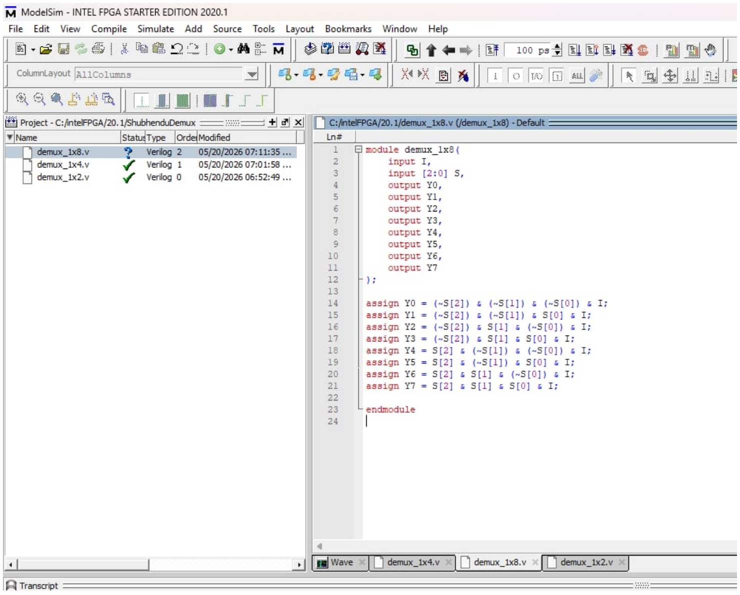
--- Simulation Waveform Output Trace ---



EXPERIMENT 11: Implementation of DEMUX 1X8

Objective: To design, simulate, and verify a 1x8 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of eight outputs.

--- Verilog Hardware Description Code ---



--- Simulation Waveform Output Trace ---